Group Theory Cheat Sheet

Group Axioms

A group is an ordered pair $(G, *)$ where G is a set and $*$ is a binary operation on G satisfying the following axioms:

- **Associativity:** $(a * b) * c = a * (b * c) \forall a, b, c \in G$
- **Identity:** $\exists e \in G$, called an identity element of G, s.t. $\forall a \in G$ we have $a * e = e * a = a$.
- **Inverse** $\forall a \in G \exists\, a^{-1} \in G$, called an inverse of a, s.t. $a * a^{-1} = a^{-1} * a = e$.

Closure is guaranteed due to the definition of binary operation.
Identity and inverses are unique.

Some Special Groups

- A group is called **abelian** if it is commutative.
- The group of all symmetries of a $n-$sided regular polygon is called the **dihedral group**. It is represented as,

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

- The group of all bijections on a set on n elements is called the **symmetric** group denoted as S_n.
- The **Klein-4** group is a group with 4 elements in which each element is a self inverse.

Homomorphisms and Isomorphisms

Let $(G, *)$ and $(H, \circ)$ be groups. A map $\varphi : G \to H$, $s.t.$ $\varphi(x * y) = \varphi(x) \circ \varphi(y)\ \forall\ x, y \in G$ is called a **homomorphism.**
A bijective homomorphism is called an **isomorphism**.

Subgroups

For a Group G a subset $H \subseteq G$, is a **subgroup** of G, i.e. $H \leq G$ if it is non empty and a group with the binary operation of G restricted to H.
Alternatively, a subset of a group is a subgroup if it is non empty and closed under products and inverses, i.e. for $H \subseteq G$

- $H \neq \emptyset$
- $xy^{-1} \in H, \forall x, y \in H$

A subgroup N of G is called **normal**, denoted as $N \trianglelefteq G$ if $gng^{-1} \in N, \forall\ g \in G, n \in N$.

Group Actions

A **group action** of a group $(G, *)$ on a set X is a map $\circ : G \times X \to X$ satisfying the following properties,

- **Identity:** $e \circ x = x, \forall x \in X$
- **Compatibility:** $g \circ (h \circ x) = g * h \circ x, \forall g, h \in G, x \in X$

Alternatively a group action on a set X can be thought of as a homomorphism from G to the symmetric group of X.
The **conjugation action** is a homomorphism $\varphi_x : G \to G$ for some fixed $x \in G$ is defined as $\varphi_x(g) = xgx^{-1}$.

Stablizer, Kernel of Group Action

If G acts on a set X then we define the following,
Orbit of a $x \in X$ is the set $Gx = \{gx \in X \mid g \in G\}$. Alternatively its the equivalence class induced by the group action.
Stablizer of an element $x \in X$ is the set of elements from the group which leave x fixed, i.e. $G_x = \{g \in G \mid gx = x\}$
Kernel of a group action is the kernel of the associated group homomorphism i.e., $\{g \in G \mid gx = x, \forall x \in X\}$
An action is called **transitive** if it has only one orbit, and is called **faithful** if its kernel is trivial.
Conjugacy classes of G is the equivalence classes of G when it acts on itself with conjugation. i.e. $\{gag^{-1} \mid g \in G\}$

Centralizers and Normalizers

Centralizer of $A \subseteq G$ in G is a subset of G defined as

$$C_G(A) = \{g \in G \mid gag^{-1} = a\ \forall\ a \in A\}.$$

It is the set of all elements of G which commute with every element of A.
Center of G is the subset of G defined as

$$Z(G) = \{g \in G \mid gx = xg\ \forall\ x \in G\}.$$

It is the set of elements commutating with all the elements of G. It is the kernel of the conjugation action.
Normalizer of A in G is defined as

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

$gAg^{-1} = \{gag^{-1} \mid a \in A\}$. Note that $C_G(A) \leq N_G(A)$. The normalizer of a subset is its stabilizer under conjugation action.

Cyclic Groups and Cycle Notation

A Group H is **cyclic** if $\exists\ x \in H$ s.t. $H = \{x^n \mid n \in \mathbb{Z}\}$
For the above case we say $H = \langle x \rangle$ and that H is generated by x.

- A cyclic group can have more than one generator.
- All cyclic groups are abelian.
- If $H = \langle x \rangle$ then $|H| = |x|$, if $|H| = n < \infty$ then $x^n = 1$
- Any two cyclic groups of the same order are isomorphic.

Two-Line to Cycle notation for permutations

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 4 & 3 & 1 \end{pmatrix} = (125)(34) = (34)(125) = (34)(512) = (15)(25)(34)$$

Here, the last form is a case of 2-cycle (transposition).

Parity of Permutations and Alternating Groups

The parity of any permutation σ is given by the parity of the number of its 2-cycles (transpositions).

Alternating Groups:
An alternating group is the group of even permutations of a finite set of length n. It is denoted by A_n it's order is $\frac{n!}{2}$

Cosets and Quotient Groups

For any $N \leq G$ and any $g \in G$

- $gN = \{gn \mid n \in N\} = \{g, gh_1, gh_2 \dots\}$ and,
- $Ng = \{ng \mid n \in N\} = \{g, h_1g, h_2g \dots\}$ are called a left coset and a right coset respectively.

For a Group G and $N \trianglelefteq G$, the **quotient group** of N in G (i.e. G/N), is the set of cosets of N in G.
The definition of a normal subgroup is the same as left and right cosets being equal.

Lagrange's Theorem and some results

Lagrange's Theorem: For a finite group G and $H \leq G$,

- The order of H divides the order of G, and,
- The number of left cosets of H in G equals $\frac{|G|}{|H|}$

Some important results

- If G is a finite group and $x \in G$, then the order of x divides the order of G, and $x^{|G|} = e\ \forall\ x \in G$
- If G is a group of prime order, then G is cyclic

Cauchy's Theorem

Cauchy's Theorem: If G is a finite group and p is a prime dividing $|G|$ then G has an element of order p.

The Isomorphism Theorems

First Isomorphism Theorem:
If $\varphi : G \to H$ is a homomorphism of groups. Then $\ker\varphi \trianglelefteq G$ and, $G/\ker\varphi \cong \varphi(G)$.
Second Isomorphism Theorem:
For a group G with, $A, B \leq G$ and, $A \trianglelefteq N_G(B)$. Then $AB \leq G$, $B \trianglelefteq AB, A \cap B \trianglelefteq A$ and, $AB/B \cong A/A \cap B$
The Third Isomorphism Theorem:
For a group G with, $H, K \trianglelefteq G$ and, $H \leq K$. Then $K/H \trianglelefteq G/H$ and, $\frac{G/H}{K/H} \cong G/K$

Orbit–stabilizer Theorem

Orbit–stabilizer Theorem:
Let a finite group G act on a set X. For $x \in X$ denote its orbit by Gx and its stabilizer by G_x. Then the map $G/G_x \to Gx$, $gG_x \mapsto gx$, is a bijection, so $|Gx| = [G : G_x]$ and hence $|Gx|\,|G_x| = |G|$.

Orbit decomposition and class equation

For a finite group G acting on a finite set X we have the **orbit decomposition** $X = \bigsqcup_i Gx_i$, where x_i run over orbit representatives, so $|X| = \sum_i |Gx_i| = \sum_i [G : G_{x_i}]$. For the conjugation action of G on itself, the orbits are conjugacy classes and the fixed points are $Z(G)$. The resulting formula

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$$

(with x_i running over non-central conjugacy classes) is the **class equation** of G.

p-group actions and fixed points

Let G be a finite p-group acting on a finite set X. Write $\mathrm{Fix}(X) = \{x \in X \mid gx = x \ \forall g \in G\}$. Then

$$|X| \equiv |\mathrm{Fix}(X)| \pmod p.$$

In particular, if $p \nmid |X|$, the action has a fixed point. Applying this to the conjugation action of a p-group on itself gives the class equation $|G| = |Z(G)| + pk$, so $Z(G) \neq 1$ and more generally every finite p-group has nontrivial center.

Cayley's Theorem

Cayley's Theorem:
Every group is isomorphic to a subgroup of some symmetric group. If G is a group of order n, then G is isomoprhic to a subgroup of S_n

Automorphisms

An **automorphism** of G is an isomorphism $G \to G$. The set $\mathrm{Aut}(G)$ is a group under composition. For $g \in G$, the map $x \mapsto gxg^{-1}$ is an **inner automorphism**. The inner automorphisms form a normal subgroup $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$ with $\mathrm{Inn}(G) \cong G/Z(G)$. The quotient $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$ is the group of **outer automorphisms**. Some facts:

- $\mathrm{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
- For $n \neq 2, 6$, every automorphism of S_n is inner, so $\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n)$.
- S_6 has a nontrivial outer automorphism: $\mathrm{Out}(S_6) \neq 1$.

p-groups and Sylow p-groups

A **p-group** is a finite group of order p^a for some prime p and $a \geq 1$. Subgroups of G which are p-groups are called p-subgroups. If $|G| = p^a m$ with $p \nmid m$, a **Sylow p-subgroup** of G is a subgroup of order p^a. The set of Sylow p-subgroups is denoted $Syl_p(G)$.

Sylow Theorems

Let $|G| = p^a m$ with $p \nmid m$.
First Sylow Theorem: G has a subgroup of order p^a (a Sylow p-subgroup).
Second Sylow Theorem: Any two Sylow p-subgroups of G are conjugate.
Third Sylow Theorem: If n_p is the number of Sylow p-subgroups, then $n_p \equiv 1 \pmod p$ and $n_p \mid m$.
("Fourth" Sylow consequence): A Sylow p-subgroup P is normal in G iff $n_p = 1$ (equivalently, P is the unique Sylow p-subgroup).

Sylow consequences

Let G be a finite group and p a prime.

- Every p-subgroup of G is contained in some Sylow p-subgroup.
- If $H \leq G$ and P is a Sylow p-subgroup of G, then $P \cap H$ is a Sylow p-subgroup of H.
- If $H \leq G$ and $[G : H]$ is coprime to p, then any Sylow p-subgroup of H is a Sylow p-subgroup of G.
- If $[G : H]$ is a power of p, then H contains a Sylow p-subgroup of G.

Frattini subgroup and Frattini argument

The **Frattini subgroup** $\Phi(G)$ of a group G is the intersection of all maximal subgroups of G (for a finite p-group, the intersection of its maximal subgroups). It is characteristic in G.
For a finite p-group G, the quotient $G/\Phi(G)$ is an elementary abelian p-group; its dimension over $\mathbb{F}_p$ equals the minimal number of generators of G.
Frattini argument: If $N \trianglelefteq G$ and P is a Sylow p-subgroup of N, then $G = N_G(P)\, N$.

Commutators

For a group G and $x, y \in G$ call $[x, y] = x^{-1}y^{-1}xy$ the **commutator** of x and y.
For subsets $A, B \subseteq G$ define the group generated by its commutators as $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$.
Similarly $G' = [G, G]$ is the subgroup of G generated by all commutators, its called the commutator subgroup of G, or its **derived subgroup**.
The following are some useful properties of commutators,

- $xy = yx \iff [x, y] = 1$
- $H \trianglelefteq G \iff [H, G] \leq H$
- G/G' is abelian and its the largest abelian quotient of G. It is called the **abelianization** of G.
- Any homomorphism from G to an abelian group A factors through G/G' (via the natural map $G \to G/G'$).

Direct products

The **direct product** $G_1 \times G_2 \times \cdots$ of groups $G_1, G_2, \ldots$ is set of sequences $(g_1, g_2, \cdots)$ with $g_i \in G_i$ and operation $*$ defined component wise.
If H, K are normal subgroups of G with $H \cap K = 1$ then $HK \cong H \times K$

Semidirect products

External semidirect product:
Given groups H, K and a homomorphism $\varphi : K \to \mathrm{Aut}(H)$, the semidirect product $H \rtimes_\varphi K$ is the set $H \times K$ with multiplication

$$(h_1, k_1)(h_2, k_2) = (h_1\, \varphi(k_1)(h_2),\, k_1 k_2).$$

[2pt] **Internal semidirect product:**
A group G is an internal semidirect product of subgroups H, K if
• $H \trianglelefteq G$, • $HK = G$, • $H \cap K = 1$.
Then the map $H \rtimes_\varphi K \to G$, $(h, k) \mapsto hk$, is an isomorphism for a suitable $\varphi : K \to \mathrm{Aut}(H)$ given by conjugation. Different φ that differ by inner automorphisms of H give isomorphic semidirect products; only the outer class of φ matters.

Exact sequences and extensions

A **short exact sequence** of groups is

$$1 \to N \xrightarrow{i} G \xrightarrow{\pi} Q \to 1,$$

where i is injective, π is surjective and $\mathrm{im}(i) = \ker(\pi)$.
This encodes that $N \trianglelefteq G$ and $G/N \cong Q$; one says that G is an **extension** of Q by N. The sequence **splits** if there is a homomorphism $s : Q \to G$ with $\pi \circ s = \mathrm{id}_Q$. In this case $G \cong N \rtimes Q$ is a semidirect product.

Finitely generated abelian groups

Every finitely generated abelian group A is isomorphic to a direct product of cyclic groups. Equivalently, there is a decomposition (noncanonically)

$$A \cong \mathbb{Z}^r \times \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z},$$

with $r \geq 0$ and $n_1 \mid n_2 \mid \cdots \mid n_k$. The integers r and n_i are uniquely determined.

Schur-Zassenhaus Theorem

Schur-Zassenhaus Theorem:
Let G be a finite group and $N \trianglelefteq G$ with $\gcd(|N|, |G/N|) = 1$. Then there exists a subgroup $H \leq G$ such that $|H| = |G/N|$ and $G = N \rtimes H$ with $N \cap H = 1$. Any two such complements H are conjugate in G.

Simple groups and Composition Series

A group G is called **simple** if $|G| > 1$ and its only normal subgroups are 1 and G.
A sequence of subgroups as follows (called a subnormal sequence),

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_{k-1} \leq N_k = G$$

is called a **composition series** if N_{i+1}/N_i is simple, for $i = 0, \ldots k-1$. The quotient groups are called composition factors.

Zassenhaus lemma and Schreier refinement

Zassenhaus (butterfly) lemma:
Let G be a group with subgroups A, C and normal subgroups $B \trianglelefteq A$, $D \trianglelefteq C$. Then there is an isomorphism

$$\frac{(A \cap C)B}{(A \cap D)B} \cong \frac{(A \cap C)D}{(B \cap C)D}.$$

This is visualized by the "butterfly" diagram in the subgroup lattice.
Schreier refinement theorem: Any two subnormal series of a group G admit equivalent refinements (their factor groups can be refined so that the multisets of factors are isomorphic up to order). Jordan–Hölder is the special case for composition series.

Jordan–Hölder theorem

A **composition series** of a nontrivial finite group G is a subnormal series

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = G$$

with each factor N_{i+1}/N_i simple. These N_{i+1}/N_i are the **composition factors**.
Jordan–Hölder theorem: Every finite nontrivial group has a composition series, and any two composition series have the same multiset of composition factors, up to isomorphism and order.

Examples of simple groups

Basic examples of finite simple groups:

- Simple abelian groups are exactly the cyclic groups C_p of prime order p.
- For $n \geq 5$, the alternating group A_n is non-abelian and simple.
- Projective special linear groups: for a field F, $SL_n(F) = \{A \in GL_n(F) \mid \det A = 1\}$, $PSL_n(F) = SL_n(F)/Z(SL_n(F))$. Over a finite field $\mathbb{F}_q$, $PSL_n(\mathbb{F}_q)$ is simple for $n \geq 2$ except $(n, q) = (2, 2)$ and $(2, 3)$; in particular $PSL_2(q)$ is simple for all $q \geq 4$.

Free groups and presentations

A **free group** on a set X is a group $F(X)$ with an injection $X \to F(X)$ such that any map $X \to G$ into a group G extends uniquely to a homomorphism $F(X) \to G$ (universal property). Elements are reduced words in the generators and their inverses.
A **presentation** of a group is $G = \langle X \mid R \rangle$, the quotient of $F(X)$ by the normal subgroup generated by a set of relators R. Every group has some presentation; if both X and R are finite, G is **finitely presented**.
(Nielsen–Schreier) Every subgroup of a free group is free.

Solvable groups

A group G is called **solvable** if it admits a subnormal series

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

with each factor G_{i+1}/G_i abelian.
Equivalently, the **derived series**

$$G^{(0)} = G, \quad G^{(1)} = [G, G], \quad G^{(i+1)} = [G^{(i)}, G^{(i)}]$$

terminates: $G^{(n)} = 1$ for some n.

Basic properties:
• If $H \leq G$ and G is solvable, then H is solvable.
• If $\varphi : G \to K$ is a homomorphism and G is solvable, then $\varphi(G)$ and $G/\ker\varphi$ are solvable.
• If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable (extensions of solvable groups are solvable).
• A finite direct product of solvable groups is solvable.

For finite groups, G is solvable $\iff$ all composition factors of G are cyclic of prime order. Burnside: if $|G| = p^a q^b$ (p, q primes), then G is solvable. Feit–Thompson: every finite group of odd order is solvable.

Nilpotent groups

A group G is called **nilpotent** if its **lower central series**

$$G_0 = G, \quad G_{i+1} = [G_i, G]$$

terminates: $G_n = 1$ for some $n \geq 0$. The least such n is the nilpotency class.

Equivalent descriptions:
• G has a finite **central series**

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

with each factor $G_{i+1}/G_i \subseteq Z(G/G_i)$.
• (Upper central series) There is a chain

$$1 = Z_0 \subseteq Z_1 \subseteq \cdots \subseteq Z_c = G$$

with $Z_{i+1}/Z_i = Z(G/Z_i)$ for all i; this terminates iff G is nilpotent.
• For finite G: G is nilpotent $\iff$ G is the (internal) direct product of its Sylow p-subgroups $\iff$ every Sylow subgroup is normal.
• (Normalizer condition, for finite G) G is nilpotent $\iff$ every proper subgroup $H < G$ satisfies $H < N_G(H)$.

Closure properties:
• Subgroups of nilpotent groups are nilpotent.
• Homomorphic images and quotients of nilpotent groups are nilpotent.
• A finite direct product of nilpotent groups is nilpotent.

In particular, every finite p-group is nilpotent, and every nilpotent group is solvable.

Linear representations

Let G be a finite group over $\mathbb{C}$.

- A (linear) representation of G is a homomorphism $\rho : G \to GL(V)$, where V is a finite-dimensional complex vector space.
- The degree of ρ is $\dim_{\mathbb{C}} V$.
- Choosing a basis of V identifies ρ with $\rho : G \to GL_n(\mathbb{C})$ (matrix representation).
- Two representations (V, ρ) and (W, σ) are equivalent if there exists a linear isomorphism $T : V \to W$ with $T\rho(g) = \sigma(g)T$ for all g.
- Finite-dimensional representations of G over $\mathbb{C}$ are the same as finite-dimensional left modules over the group algebra $\mathbb{C}G$.

Subrepresentations and irreducibility

Let (V, ρ) be a representation of G.

- A subspace $W \subseteq V$ is G-invariant if $\rho(g)W \subseteq W$ for all $g \in G$. Then W is a subrepresentation.
- The quotient V/W becomes a representation via $\rho(g)(v + W) = \rho(g)v + W$.
- A nonzero representation is irreducible if it has no proper nonzero subrepresentation.
- A representation is reducible if it is not irreducible.
- A representation is completely reducible if it is a direct sum of irreducible subrepresentations.
- Direct sum: if (V_i, ρ_i) are representations, then $\bigoplus_i V_i$ with $g \cdot (v_1, \ldots, v_k) = (\rho_1(g)v_1, \ldots, \rho_k(g)v_k)$ is a representation.

Basic constructions and examples

- Trivial representation: $V = \mathbb{C}$, $\rho(g) = 1$.
- Sign representation of S_n: $V = \mathbb{C}$, $\rho(\sigma) = \mathrm{sgn}(\sigma)$.
- Permutation representation: if G acts on a finite set X, let $V = \mathbb{C}^X$ with basis $\{e_x\}$ and $g \cdot e_x = e_{gx}$.
- Regular representation: $V = \mathbb{C}G$ with basis $\{e_g\}$ and $h \cdot e_g = e_{hg}$.
- Tensor product: for V, W, define $g \cdot (v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$.
- Dual representation: $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ with $(g \cdot f)(v) = f(\rho(g^{-1})v)$.

Intertwiners and Schur's Lemma

Let (V, ρ) and (W, σ) be representations.

- A homomorphism of representations (intertwiner) is a linear map $T : V \to W$ with $T\rho(g) = \sigma(g)T$ for all g.
- Denote by $\mathrm{Hom}_G(V, W)$ the space of intertwiners. Put $\mathrm{End}_G(V) = \mathrm{Hom}_G(V, V)$.
- Schur's Lemma (1): if V, W are irreducible, any $T \in \mathrm{Hom}_G(V, W)$ is either 0 or an isomorphism.
- Schur's Lemma (2): if V is irreducible over $\mathbb{C}$, then $\mathrm{End}_G(V)$ consists of scalar multiples of the identity.

Unitary representations and Maschke's theorem

- A Hermitian inner product $(\cdot, \cdot)$ on V is G-invariant if $(\rho(g)v, \rho(g)w) = (v, w)$ for all g. Then $\rho(g)$ is unitary.
- For any inner product $(\cdot, \cdot)$ on V, define $(v, w)_G = \frac{1}{|G|} \sum_{g \in G} (\rho(g)v, \rho(g)w)$. Then $(\cdot, \cdot)_G$ is G-invariant.
- Over $\mathbb{C}$, every finite group representation is equivalent to a unitary representation.
- Maschke's Theorem: if $\mathrm{char}(\mathbb{C}) = 0$ (or more generally $\mathrm{char}(F) \nmid |G|$), every representation of G is completely reducible.
- Equivalently, for every G-invariant subspace $W \subseteq V$ there exists a G-invariant complement W' with $V = W \oplus W'$.

Characters: definition and basic facts

Let (V, ρ) be a representation.

- The character of V is $\chi_V : G \to \mathbb{C}$ given by $\chi_V(g) = \mathrm{tr}(\rho(g))$.
- $\chi_V(1) = \dim_{\mathbb{C}} V$.
- If $V \cong W$ as representations, then $\chi_V = \chi_W$.
- For irreducible representations over $\mathbb{C}$, equality of characters implies equivalence.
- χ_V is constant on conjugacy classes (class function).
- For direct sums and tensor products: $\chi_{V \oplus W} = \chi_V + \chi_W$, and $\chi_{V \otimes W} = \chi_V \chi_W$ (pointwise).
- If V is unitary, then $\chi_{V^*}(g) = \overline{\chi_V(g)}$.
- For a permutation representation from an action on X, $\chi(g)$ equals the number of fixed points of g in X.

Inner product and orthogonality of characters

Let $\mathrm{CF}(G)$ be the space of complex class functions on G.

- Inner product on $\mathrm{CF}(G)$: $\langle \varphi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \varphi(g)\overline{\psi(g)}$.
- If χ_V, χ_W are characters, then $\langle \chi_V, \chi_W \rangle$ is a nonnegative integer. It equals the multiplicity of W in V.
- Orthogonality: if χ_i, χ_j are irreducible characters, then $\langle \chi_i, \chi_j \rangle = \delta_{ij}$.
- A character χ is irreducible iff $\langle \chi, \chi \rangle = 1$.
- If $V \cong \bigoplus_i m_i V_i$ with V_i irreducible, then $\chi_V = \sum_i m_i \chi_i$ and $m_i = \langle \chi_V, \chi_i \rangle$.

Main character formulas

Let $C_1, \ldots, C_h$ be the conjugacy classes of G and $\chi_1, \ldots, \chi_h$ the irreducible characters.

- The number of irreducible characters equals the number of conjugacy classes.
- The irreducible characters form an orthonormal basis of $\mathrm{CF}(G)$ with respect to the inner product above.
- Let $d_i = \chi_i(1)$ be the degrees of irreducible representations. Then $\sum_i d_i^2 = |G|$.
- Row orthogonality: for $i \neq j$, $\sum_{g \in G} \chi_i(g)\overline{\chi_j(g)} = 0$, and for $i = j$, the sum is $|G|$.
- Column orthogonality: if g, h are representatives of different conjugacy classes, then $\sum_i \chi_i(g)\overline{\chi_i(h)} = 0$; if g, h lie in the same class C, then $\sum_i \chi_i(g)\overline{\chi_i(h)} = |G|/|C|$.

Regular representation and group algebra

Let R be the regular representation on $V = \mathbb{C}G$.

- The character of R satisfies $\chi_R(1) = |G|$ and $\chi_R(g) = 0$ for $g \neq 1$.
- If V_i are the irreducible representations with degrees d_i, then $R \cong \bigoplus_i d_i V_i$.
- Comparing characters gives $\sum_i d_i^2 = |G|$ and shows that V_i occurs in R with multiplicity d_i.
- As an algebra, $\mathbb{C}G \cong \prod_i M_{d_i}(\mathbb{C})$.
- There exist central idempotents $e_i \in \mathbb{C}G$ with $1 = \sum_i e_i$, $e_i e_j = 0$ for $i \neq j$, and $e_i \mathbb{C}G \cong M_{d_i}(\mathbb{C})$.

Permutation characters and Burnside's lemma

Let G act on a finite set X and consider the permutation representation on $V = \mathbb{C}^X$.

- The character satisfies $\chi(g) = |\{x \in X \mid gx = x\}|$.
- The number of G-orbits in X is $\frac{1}{|G|} \sum_{g \in G} \chi(g)$. This is the Cauchy–Frobenius (Burnside) lemma.
- For the conjugation action of G on itself, permutation characters encode the sizes of conjugacy classes and centralizers.

Restriction, induction and Frobenius reciprocity

Let $H \leq G$ be a subgroup.

- Restriction: a representation V of G becomes a representation of H by restricting ρ to H. For characters: $(\mathrm{Res}_H^G \chi)(h) = \chi(h)$ for $h \in H$.
- Induction: a representation W of H has an induced representation $\mathrm{Ind}_H^G W$ of G with $\dim(\mathrm{Ind}_H^G W) = [G : H] \dim W$.
- Frobenius reciprocity (characters): if θ is a character of H and χ a character of G, then $\langle \mathrm{Ind}_H^G \theta, \chi \rangle_G = \langle \theta, \mathrm{Res}_H^G \chi \rangle_H$.

Irreducible representations of abelian groups

Let G be a finite abelian group.

- Every irreducible representation of G over $\mathbb{C}$ is 1-dimensional.
- Irreducible representations are group homomorphisms $\chi : G \to \mathbb{C}^\times$.
- There are exactly $|G|$ distinct irreducible characters; they form the dual group $\widehat{G}$ under pointwise product.
- The regular representation of G decomposes as the direct sum of all irreducible characters, each with multiplicity 1.
- The character table of G is a square matrix of size $|G| \times |G|$ and is a discrete Fourier transform on G.

Example: representations of S_3

For S_3:

- Conjugacy classes: identity (1), transpositions (12), 3-cycles (123). Class sizes: $1, 3, 2$.
- There are 3 irreducible representations:
  - Trivial: degree 1, character values $(1, 1, 1)$.
  - Sign: degree 1, character values $(1, -1, 1)$.
  - Standard 2-dimensional: degree 2, character values $(2, 0, -1)$.
- Degrees satisfy $1^2 + 1^2 + 2^2 = 6 = |S_3|$.
- The regular representation of S_3 decomposes as $R \cong 1 \oplus 1_{\text{sgn}} \oplus 2 \cdot V_{\text{std}}$.